Sentinel-3A two-month OEM output report

Purpose and scope

This report explains the archived orbit predictions exported as CCSDS Orbit Ephemeris Messages (OEM). The export preserves the completed experimental results; it does not run new orbit determination, improve the model, reconstruct missing forecasts, or substitute reference orbit positions for predictions.

The files contain daily benchmark samples, not dense orbit tracks. They are appropriate for reviewing the saved one-day through seven-day predictions. Their 24-hour spacing cannot resolve Sentinel-3A's roughly 100-minute orbit. Do not interpolate them for tracking, passes, continuous ground tracks or operational use. A dense ephemeris requires another propagation/export at a suitable cadence.

Which file should be presented?

File or folder	Meaning
improved-broadcast-all-issued.oem	The newer nominal Broadcast-only estimation trial: 60 independent forecasts. Start with this file when presenting the improved Broadcast experiment.
original-four-tier-all-issued.oem	Original comparison using Final, Rapid, Ultra-Rapid and Broadcast GPS products: 205 forecasts, with product tier in each segment's comments.
retrospective-old-maneuvers-all-issued.oem	60 replays using the newly estimated state and the original candidate maneuver reconstruction. Diagnostic and retrospective.
retrospective-corrected-maneuvers-all-issued.oem	60 replays using background-corrected candidate maneuvers. Diagnostic and retrospective.
Group folders	One OEM per issued run, suitable for readers expecting a single segment.
coverage.csv	Every attempted case, including rejected cases with no OEM.
manifest.json	Output hashes and the paths/hashes of the original CSV sources.

The two maneuver files are separate experimental profiles. They are not corrections that should automatically replace all nominal predictions. Their future, reference-derived maneuver information was unavailable at the original cutoff. No per-date best-performing profile was chosen to construct this package.

Time window and completeness

- Observation campaign daily forecast cutoffs: **3 September–2 November 2024**, inclusive: 61 dates at **05:59:42 UTC**. - Each issued run has a seven-day horizon and eight state samples: elapsed days 0, 1, 2, 3, 4, 5, 6 and 7. - The first forecast spans 3–10 September; the last spans 2–9 November. - Overall exported sample epochs span **3 September–9 November 2024**. - These are rolling forecasts; they are not one unbroken two-month propagation.

Experiment	Attempted cases	Issued OEMs	Not issued
Original four-tier comparison	244	205	39
Improved Broadcast	61	60	1
Replay: original candidate maneuvers	61	60	1
Replay: background-corrected maneuvers	61	60	1
Total	427	385	42

The improved Broadcast run for **16 October 2024** failed the fit-quality gate; its replay profiles also have no issued output. All original comparison rejections remain visible in the coverage table. A missing OEM is a rejected/unissued result, not zero error. Availability of a prediction through 9 November does not imply reference data exists to score every late prediction endpoint.

OEM format and structure

The output is **CCSDS OEM version 2.0**, **KVN** (Keyword = Value Notation), stored as ASCII text. OEM is a standard way to exchange sampled orbital position and velocity. It is not a TLE, a raw GPS observation file, a broadcast navigation message, a force-model configuration, or an error report.

A file has a header and one or more metadata/data segments. A representative single-run extract is:

```
CCSDS_OEM_VERS = 2.0
CREATION_DATE = 2026-11-12T04:11:44
```

COMMENT Campaign: improved-broadcast; case: 2024-09-03-broadcast COMMENT UTC cutoff: 2024-09-03T05:59:42.000 META_START OBJECT_NAME = SENTINEL-3A OBJECT_ID = 2016-011A CENTER_NAME = EARTH REF_FRAME = ITRF TIME_SYSTEM = UTC START_TIME = 2024-09-03T05:59:42.000 STOP_TIME = 2024-09-10T05:59:42.000 META_STOP COMMENT Position km; velocity km/s. No interpolation method asserted. 2024-09-03T05:59:42.000 2991.737690361976 5993.733816487692 2585.138768381534 2.696252010539 1.624476786231 -6.858906667896 ```

The extract omits additional provenance comments and the remaining seven data lines for readability. The actual files preserve all those comments and samples.

Header and metadata fields

Field	Explanation
`CCSDS_OEM_VERS`	OEM format version, 2.0.
`CREATION_DATE`	UTC time of this export, not the observation date, forecast cutoff or original experiment execution time. A 2026 export can describe a 2024 historical experiment.
`ORIGINATOR`	Producing project: URSC.
`COMMENT`	Human-readable provenance, profile, cutoff and caveats; not another state sample. Preserve these when sharing files.
`META_START` / `META_STOP`	Delimit metadata for one independent segment. The state lines follow `META_STOP`.
`OBJECT_NAME`	Satellite being predicted: SENTINEL-3A.
`OBJECT_ID`	International designator: 2016-011A.
`CENTER_NAME`	Earth is the coordinate origin.
`REF_FRAME`	ITRF, an Earth-fixed rotating terrestrial frame. This export retains the archived frame rather than transforming again. It does not add a claim of a particular ITRF realization.
`TIME_SYSTEM`	UTC applies to state epochs and segment bounds. These timestamps do not use GPS time.
`START_TIME` / `STOP_TIME`	First and last sample in that segment, not the bounds of the whole input archive.

No `INTERPOLATION` or `INTERPOLATION_DEGREE` is asserted: daily samples must not be made to look like a continuous high-fidelity orbit. No covariance blocks are present because the source campaign CSVs did not store covariance.

Reading a state line

Each state line contains seven whitespace-separated fields:

```
epoch_UTC X_km Y_km Z_km VX_km/s VY_km/s VZ_km/s
```

For the illustrated state:

Component	Value	Meaning
Epoch	2024-09-03 05:59:42 UTC	Day-zero cutoff state for this forecast.
X	2991.737690361976 km	Earth-fixed Cartesian X position.
Y	5993.733816487692 km	Earth-fixed Cartesian Y position.
Z	2585.138768381534 km	Earth-fixed Cartesian Z position.
VX	2.696252010539 km/s	X velocity expressed consistently in the rotating frame.
VY	1.624476786231 km/s	Y velocity expressed consistently in the rotating frame.
VZ	-6.858906667896 km/s	Z velocity expressed consistently in the rotating frame.

X points toward the equator at the Greenwich meridian, Y toward the equator at 90 degrees east, and Z toward the terrestrial north pole. These Cartesian coordinates are not latitude, longitude and altitude. Position norm is distance from Earth's centre, not height above the ellipsoid. A negative component means the vector points opposite that axis; it is not an error or a negative altitude.

ITRF velocity must not be treated as inertial velocity. Converting position and velocity to GCRF requires the appropriate epoch-dependent Earth-orientation transform, including the rotating-frame velocity term. The exporter copies the existing ITRF position and velocity and does not perform a second transformation.

The source CSV uses metres and metres per second. Every component is divided by 1000 for OEM. For example, 2,991,737.690361976 m becomes 2991.737690361976 km. Twelve fractional digits preserve the stored decimal values; they do not imply picometre physical accuracy.

Why combined files overlap

Each successful cutoff is an independently initialized seven-day prediction. The 4 September segment overlaps the 3 September segment over most of their horizons. Two positions at the same UTC can therefore differ: they belong to different forecast issue dates, horizons or product profiles.

The combined file contains separate metadata/data segments and preserves each forecast. It does not average, overwrite or concatenate their state lines into one chronological track. Some OEM consumers cannot accept overlapping segments; use the individual files with those tools. Within each segment, epochs increase strictly and have 86,400-second spacing.

For accuracy assessment, compare a sample to the independent reference at the same UTC, in the same frame and units, and keep its forecast issue date. Day one means one day after that segment's cutoff; day seven means seven days after it. Do not compare a day-one state from one segment with a day-seven state from another and call them the same forecast horizon.

Forecast provenance and interpretation

The nominal workflow uses pre-cutoff GNSS observations to form position fixes, fits an initial orbit, and propagates it with the experiment's force assumptions. The stored future weather/EOP policy is retained by this export. It does not inject new measured future weather, precise reference states or new attitude information.

However, the original Final and Rapid GPS products were retrospective products and may incorporate post-cutoff information. Ultra-Rapid publication eligibility was not verified. The archived experiments do not establish publication-vintage eligibility for every input, including the Broadcast trial. Consequently these exports must not be presented as a fully verified operational as-issued forecast campaign. The forecast/reconstruction distinction survives the format conversion.

The maneuver replays use unconfirmed events inferred from future reference orbits. They measure a retrospective compensation experiment; they do not demonstrate that those events could have been forecast or that official maneuver telemetry was available. All such files are labelled `retrospective` in names and comments.

OEM values alone do not establish the one-day 10 m or seven-day 1 km targets. Errors need independent reference comparisons; the absence of covariance is also not a claim of zero uncertainty. Previous performance reports remain the source for scored accuracy, and late endpoints without reference coverage remain unscored.

Validation performed

1. Exported all 385 issued source CSVs; kept all 427 attempts in the coverage table. 2. Checked eight samples per issued forecast, finite state values, increasing epochs, 86,400-second spacing, frame and bounds. 3. Compared all six state components with their source using exact decimal arithmetic after reversing the unit conversion. No value was altered. 4. Parsed all 389 OEM files with the project's existing Java OEM reader: 385 individual files plus four aggregate files. 5. Recorded source/output SHA-256 hashes and verified the ZIP archive integrity.

There are **3,080 unique sampled state records**. Aggregates duplicate the individual records, so parsing all files counts 6,160 records and 770 segments. These checks establish export fidelity and project-parser compatibility. They are not an independent external CCSDS conformance certification, a physical accuracy validation, or a certification for operational use.

Package and reproducibility

Download the complete package. It includes this report, the OEMs, coverage, provenance manifest, verification summary and checksums. The Markdown report is intended for GitHub reading; the companion PDF is intended for submission and printing.

`export_oem.py` converts the archived campaign CSVs when those source directories are available. It does not acquire missing observations or regenerate experiments. Some source directories may exist only in the research workspace; the manifest records exact CSV hashes so source identity remains checkable. Downloaded OEMs can be read without those source files or a Java installation.

When sharing with the guide, include this report and the profile being discussed. For a new dense run, specify the cutoff campaign, model profile, sampling interval, reference frame and forecast policy before generating another output set.